

## Confidence Interval

Illustrated through inference on one population mean or proportion

### Motivation & simple random sample

Eg) We wish to estimate the average height of adult US males

➔ Take a random sample.

- “Simple” random sample: every subject in the population has the same chance to be selected.

### Introduction to statistical inference on one population mean

For a “random sample” of size  $n$ :  $X_1, X_2, \dots, X_n$

<i> Point estimator  $\bar{X} \rightarrow$  sample mean  $(= \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{\sum_{i=1}^n X_i}{n})$

Other estimators: median, mode, trimmed mean, ...

### <ii> Confidence Interval (C.I.)

Eg) 95% C.I. for  $\mu$

99.9999% C.I. (‘6-9’ in the manufacture industry)

### <iii> Hypothesis Test

Eg)  $H_0: \mu \leq 5'6''$

$H_1: \mu > 5'6''$

Point Estimator, C.I., Test  $\Rightarrow$  Statistical Inference

- Draw some conclusion on the population (parameters of interest) based on a random sample.

# 1. The Exact Confidence Interval for $\mu$ when the population is normal & $\sigma^2$ is known

## ① Point estimator and confidence interval for $\mu$

- When the population is normal and the population variance is known.
- Let  $X_1, X_2, \dots, X_n$  be a random sample for a normal population with mean  $\mu$  and variance  $\sigma^2$ . That is,  $X_i \stackrel{iid.}{\sim} N(\mu, \sigma^2), i = 1, \dots, n$ .
- For now, we assume that  $\sigma^2$  is known.

<i> Point Estimator for  $\mu$  :  $\hat{\mu} = \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$E(\hat{\mu}) = E(\bar{X}) = \mu \Rightarrow \hat{\mu} = \bar{X}$  is an unbiased estimator of  $\mu$

- Intuitively, this means if you take “many” samples of size  $n$  from the population, then the mean of these samples means would be equal to  $\mu$  if you take a large enough # of samples.
- $\bar{X}$  is also a maximum likelihood estimator (MLE) of  $\mu$ .
- $\bar{X}$  is also a method of moment estimator (MOME) of  $\mu$ .
- Other good properties too.

<ii> Confidence Interval for  $\mu$

- Intuitive approach (backwards derivation for the CI boundaries  $C_1$  and  $C_2$ ):

$$P(C_1 \leq \mu \leq C_2) = 0.95$$

$$P(-C_1 \geq -\mu \geq -C_2) = 0.95$$

$$P(\bar{X} - C_1 \geq \bar{X} - \mu \geq \bar{X} - C_2) = 0.95$$

$$P\left(\frac{\bar{X} - C_1}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - C_2}{\sigma/\sqrt{n}}\right) = 0.95$$

Since we know

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

We can compute the expressions for  $C_1$  and  $C_2$ .

However, one question is that there are MANY ways to choose the C's.

Later you will see that for pivotal quantity with symmetric pdfs, the symmetric CIs are the optimal – in that they have the shortest lengths for the given confidence level  $100(1-\alpha)\%$ .

**Now we present a general approach to derive the CI's.**

**General approach for deriving CI's : the Pivotal Quantity (P.Q.) approach**

\*Definition: A pivotal quantity is a function of the sample and the parameter of interest. Furthermore, its distribution is entirely known.

1. We start by looking at the point estimator of  $\mu$ .  $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

\* Is  $\bar{X}$  a pivotal quantity for  $\mu$ ?

→  $\bar{X}$  is not because  $\mu$  is unknown.

\* function of  $\bar{X}$  and  $\mu$  :  $\bar{X} - \mu \sim N(0, \frac{\sigma^2}{n})$

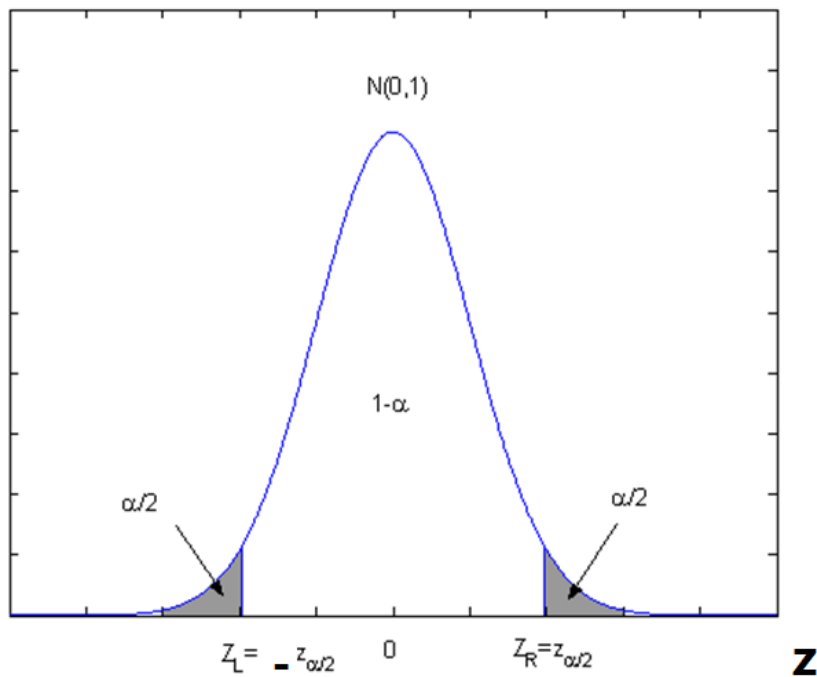
→ Yes, it is pivotal quantity.

\* Another function of  $\bar{X}$  and  $\mu$  :  $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$

→ Yes, it is pivotal quantity.

So, Pivotal Quantity is not unique.

2. Now that we have found the pivotal quantity Z, we shall start the derivation for the symmetrical CI's for  $\mu$  from the PDF of the pivotal quantity Z



100(1- $\alpha$ )% CI for  $\mu$ ,  $0 < \alpha < 1$

(e.g.  $\alpha = 0.05 \Rightarrow 95\%$  C.I.)

$$P(-Z_{\alpha/2} \leq Z \leq Z_{\alpha/2}) = 1 - \alpha$$

$$P\left(-Z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(-Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$P\left(-\bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq -\mu \leq -\bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$P\left(\bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \geq \mu \geq \bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$P\left(\bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$\therefore$  the 100(1- $\alpha$ )% C.I. for  $\mu$  is  $\left[\bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right]$

**\*Note, some special values for  $\alpha$  and the corresponding  $Z_{\alpha/2}$  values are:**

**1. The 95% CI, where  $\alpha = 0.05$  and the corresponding  $Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96$**

2. The 90% CI, where  $\alpha = 0.1$  and the corresponding  $Z_{\frac{\alpha}{2}} = Z_{0.05} = 1.645$

3. The 99% CI, where  $\alpha = 0.01$  and the corresponding  $Z_{\frac{\alpha}{2}} = Z_{0.005} = 2.575$

**Example 1.** A random sample of 400 adult US male was taken and the sample mean was found to be  $\bar{X} = 5'7'' = 67 \text{ inches}$ . Based on past studies, it is believed that the population distribution of all adult US male is normal and the standard deviation is 30 inches. Please construct a 95% confidence interval for the average height of all adult US male based on this sample.

**Solution:** The 95% CI for  $\mu$  is

$$\left[ \bar{X} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \right] = \left[ 67 - 1.96 \frac{30}{\sqrt{400}}, 67 + 1.96 \frac{30}{\sqrt{400}} \right] \approx [64, 70]$$

That is, the estimated 95% confidence interval for the average height of all adult US male is [5'4", 5'10"].

...

...

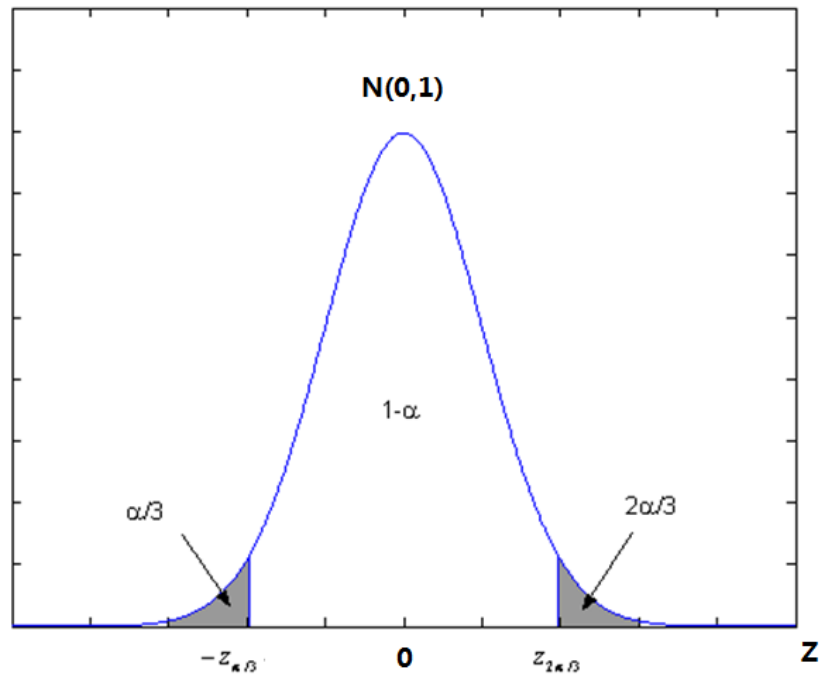
This means that we are 95% sure the population mean  $\mu$  would lie between 5'4" and 5'10".

∴ Recall the  $100(1-\alpha)\%$  symmetric C.I. for  $\mu$  is  $\left[ \bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right]$

**\*Please note that this CI is symmetric around  $\bar{X}$**

**The length of this CI is:**  $L_{sy} = 2 \cdot Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

Now we derive a non-symmetrical CI:



$$P(-Z_{\alpha/3} \leq Z \leq Z_{2/3\alpha}) = 1 - \alpha$$

100(1- $\alpha$ )% C.I. for  $\mu$

$$\Rightarrow [\bar{X} - Z_{2/3\alpha} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{1/3\alpha} \cdot \frac{\sigma}{\sqrt{n}}]$$

Compare the lengths of the C.I.'s, one can prove theoretically that:

$$L = (Z_{\alpha/3} + Z_{2/3\alpha}) \cdot \frac{\sigma}{\sqrt{n}} > L_{sy} = 2 \cdot Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

You can try a few numerical values for  $\alpha$ , and see for yourself. For example,

$$\alpha = 0.05$$

HW: Please derive the 100(1- $\alpha$ )% symmetric C.I. for  $\mu$  based on a random sample from a normal population with unknown variance

## 2. (Large Sample) Confidence interval for a population mean (\*any population) or a population proportion p

<Theorem> Central Limit Theorem

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \xrightarrow{n \rightarrow \infty} N(0,1)$$

When n is large enough, we have

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

That means Z follows approximately the normal (0,1) distribution.

**Application #1. Inference on  $\mu$  when the population distribution is unknown but the sample size is large**

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

By **Slutsky's Theorem** We can also obtain another pivotal quantity when  $\sigma$  is unknown by plugging the sample standard deviation S as follows:

$$Z = \frac{\bar{X} - \mu}{S / \sqrt{n}} \sim N(0,1)$$

We subsequently obtain the  $100(1-\alpha)\%$  C.I. using the second P.Q. for  $\mu$ :

$$\bar{X} \pm Z_{\alpha/2} \frac{S}{\sqrt{n}}$$

**Application #2. Inference on one population proportion p when the population is**

**Bernoulli(p) \*\*\*** Let  $X_i \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$ ,  $i = 1, \dots, n$ , please find the  $100(1-\alpha)\%$  CI for p.

Point estimator :  $\hat{p} = \bar{X} = \frac{\sum_{i=1}^n X_i}{n}$  (ex.  $n=1000$ ,  $\hat{p} = 0.6$ )

Our goal: derive a  $100(1-\alpha)\%$  C.I. for p

Thus for the Bernoulli population, we have:

$$\mu = E(X) = p$$

$$\sigma^2 = \text{Var}(X) = p(1-p)$$

Thus by the CLT we have:

$$Z = \frac{\bar{X} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$$

Furthermore, we have for this situation:  $\bar{X} = \hat{p}$

Therefore we obtain the following pivotal quantity Z for p:

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$$

By Slutsky's theorem, we can replace the population proportion in the denominator with the sample proportion and obtain another pivotal quantity for p:

$$Z^* = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim N(0,1)$$

**# Thus the  $100(1-\alpha)\%$  (approximate, or large sample) C.I. for p based on the second pivotal quantity  $Z^*$  is:**

$$P(-z_{\alpha/2} \leq Z^* \leq z_{\alpha/2}) = 1 - \alpha$$

$$P\left(-z_{\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(-\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq -p \leq -\hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) = 1 - \alpha$$

$$P\left(\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) = 1 - \alpha$$

=> The  $100(1-\alpha)\%$  large sample C.I. for p is

$$\left[ \hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right].$$

# CLT => n large usually means  $n \geq 30$

# special case for the inference on p based on a Bernoulli population. The sample size n is large means

Let  $X = \sum_{i=1}^n X_i$ , large sample means:

$n\hat{p} = X \geq 5$  (\*Here X= total # of 'S'), and

$n(1-\hat{p}) = n - X \geq 5$  (\*Here n-X= total # of 'F')



### Example 2.

During one of the “beer wars” in the early 1980’s, a taste test between Schlitz and Budweiser was the focus of a TV commercial. 100 people agreed to drink 2 unmarked mugs and indicate which of the two beers they liked better. 54 chose “Bud”. Construct and interpret the corresponding 95% confidence interval for  $p$  - the proportion of beer drinkers who prefer Bud to Schlitz.

### Solution.

#### Confidence Interval for one population proportion (p) when the sample size is large

Sample size :  $n$  ( $n = 100$ )

$$\text{Sample proportion : } \hat{p} = \frac{\sum_{i=1}^n X_i}{n} \quad (\hat{p} = \frac{54}{100})$$

\*\*\* Recall we usually denote  $X = \sum_{i=1}^n X_i$

“sample is large” means

- For one population mean,  $n \geq 30$
- For one population proportion :  $X \geq 5$  and  $(n - X) \geq 5$   
( $X = 54 \geq 5$  ;  $n - X = 46 \geq 5$ )

$n = 100$ ,  $X = 54$ , 95% CI for  $p$

From 95% confidence interval,  $1 - \alpha = 0.95$ ,  $\alpha = 0.05$ ,  $\frac{\alpha}{2} = 0.025$

$$\hat{p} = \frac{54}{100} = 0.54 ; Z_{0.025} = 1.96$$

$$\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{(0.54)(0.46)}{100}} = 0.049$$

$$Z_{0.025} \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = 1.96 \times 0.049 = 0.096$$

$\therefore$  The 95% confidence interval for  $p$  is  $[0.444, 0.636]$

If  $n = 10000$  ;  $\hat{p} = 0.54$ ,

$$\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{(0.54)(0.46)}{10,000}} = 0.0049$$

$$Z_{0.025} \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = 1.96 \times 0.0049 = 0.0096 \approx 0.01$$

$\therefore$  The 95% confidence interval for  $p$  is  $[0.53, 0.55]$

### 3. The Exact Confidence Interval for $\mu$ when the population is normal & $\sigma^2$ is unknown

1. Point estimation :  $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
2.  $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$
3. **Theorem.** Sampling from normal population
  - a.  $Z \sim N(0,1)$
  - b.  $W = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$
  - c.  $Z$  and  $W$  are independent.

**Definition.**  $T = \frac{Z}{\sqrt{W/(n-1)}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$

----- Derivation of CI, normal population,  $\sigma^2$  is unknown -----

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$  is not a pivotal quantity.

$\bar{X} - \mu \sim N(0, \frac{\sigma^2}{n})$  is not a pivotal quantity.

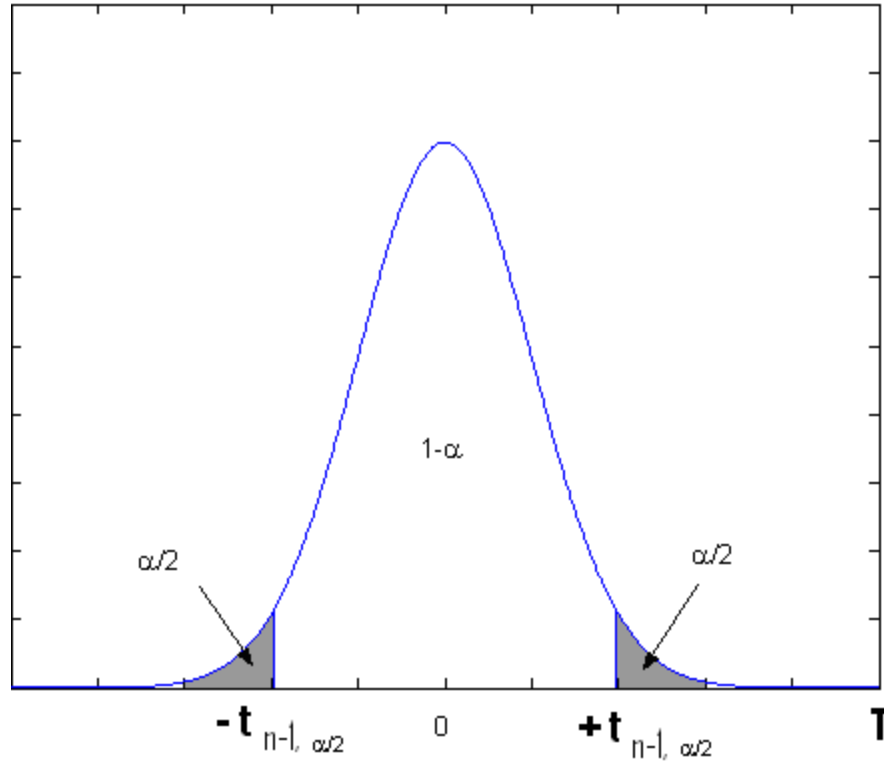
$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$  is not a pivotal quantity.

Remove  $\sigma$  !!!

Therefore  $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$  is a pivotal quantity.

Now we will use this pivotal quantity to derive the  $100(1-\alpha)\%$  confidence interval for  $\mu$ .

We start by plotting the pdf of the t-distribution with  $n-1$  degrees of freedom as follows:



The above pdf plot corresponds to the following probability statement:

$$P(-t_{n-1, \alpha/2} \leq T \leq t_{n-1, \alpha/2}) = 1 - \alpha$$

$$\Rightarrow P\left(-t_{n-1, \alpha/2} \leq \frac{\bar{X} - \mu}{S / \sqrt{n}} \leq t_{n-1, \alpha/2}\right) = 1 - \alpha$$

$$\Rightarrow P\left(-t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \leq \bar{X} - \mu \leq t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(-\bar{X} - t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \leq -\mu \leq -\bar{X} + t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\bar{X} + t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \geq \mu \geq \bar{X} - t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\bar{X} - t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$\Rightarrow$  Thus the  $100(1-\alpha)\%$  C.I. for  $\mu$  when  $\sigma^2$  is unknown is

$$\left[\bar{X} - t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}\right].$$

(\*Please note that  $t_{n-1, \alpha/2} \geq Z_{\alpha/2}$ )

**Example 3.** In a psychological **depth-perception** test, a random sample of  $n=14$  airline pilots were asked to judge the distance between 2 markers at the other end of a laboratory. The data (in test) are

2.7, 2.4, 1.9, 2.4, 1.9, 2.3, 2.2, 2.5, 2.3, 1.8, 2.5, 2.0, 2.2, **2.6**

Please construct a 95% CI for  $\mu$ , the average distance.

**Solution.**

*(Note: we can perform the Shapiro-Wilk test to examine whether the sample comes from a normal population or not. This test is not required in our class. Here we simply assume the population is normal. I will always give you such information in the exams.)*

CI for  $\mu$ , small sample, normal population, population variance unknown.

$n=14$ ,  $\bar{X} = 2.26$ ,  $S = 0.28$ ,  $\alpha=0.05$

95% CI for  $\mu$  is  $\bar{X} \pm t_{n-1, \alpha/2} \cdot \frac{S}{\sqrt{n}} = 2.26 \pm 2.16 \cdot \frac{0.28}{\sqrt{14}}$

$\therefore [2.10, 2.42]$